# INFINITY TUTORS

# PHY 122

## ELECTRICITY AND MAGNETISM, VIBRATIONS AND WAVES

## TUTORIAL 4.

ELECTRIC POTENTIAL

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## Instructions

1. Answer all the questions
2. Show your work clearly
3. Verify your work with the Infinity Tutors

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## SOME IMPORTANT NOTES

# THE ELECTRIC POTENTIAL

**Electric Potential** $V$ **at a point** ( or simply **Potential** ) *is defined as the potential energy per unit charge at a point in an electric field.*

$$V = \frac{U}{q}$$

where $U$ is electric potential energy and $q$ is charge.
Note that electric potential is a scalar, not a vector.

***Electric potential between any two points i and f in an electric field*** *is equal to the difference in potential energy per unit charge between the two points:*

$$\boxed{\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q}}$$

*We know that* $\Delta U = -W$ *where W is work done on the charge; we can substitute for* $\Delta U$ *so potential difference is;*

$$\Delta V = V_f - V_i = \frac{-W}{q}$$

*If we set* $U_i = 0$ *at infinity as our reference potential energy, then by above equation, the electric potential V must also be zero there.*
*Then from equation above, we can define the electric potential at any point in an electric field to be*

$$\Delta V = V_f - V_i = \frac{-W_\infty}{q}$$

*The SI unit for electric potential* ***Volts*** *abbreviated as* $V$*; Thus*

$$1\ Volt = 1\ joule\ per\ coulomb$$

***One electron-volt (eV)*** *is the energy equal to the work required to move a single elementary charge e, such as that of the electron or the proton, through a potential difference of exactly one volt. This tells us that the magnitude of this work is* $q\Delta V$*; so*

$$\begin{aligned} 1\,eV &= e \times 1\,V \\ &= (1.6 \times 10^{-19} C) \times 1\,\frac{J}{C} \\ &= 1.6 \times 10^{-19}\,J. \end{aligned}$$

1. **The electric potential V due to a particle of charge q at any radial distance r from the particle.**

$$V = ke\frac{q}{r}$$

2. **The electric Potential Due to Group of Point Charges**
   *We can find the net potential at a point due to a group of point charges with the help of the superposition principle (**Algebraic Sum**). The sign of the charge included, we calculate separately the potential resulting from each charge at the given point.Then we sum the potentials. For n charges, the net potential is*

$$V = \sum_{i=1}^{n} V_i = V_1 + V_2 + V_3 + \ldots + V_n = ke\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \ldots \frac{q_n}{r_n}\right)$$

# 1 ELECTRIC FIELD

1. What is the electric potential at point P, located at the center of the square of point charges shown in **Figure** 1? The distance d is 1.3 m, and the charges are [7]

$$q_1 = +12\,nC \qquad q_3 = +31\,nC$$
$$q_2 = -24\,nC \qquad q_4 = +17\,n$$

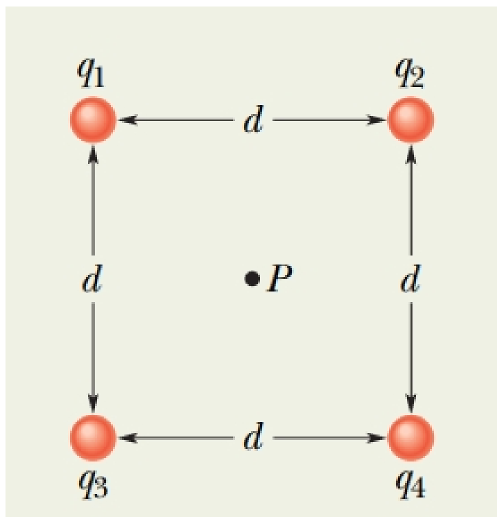


Figure 1: Four point charges are held fixed at the corner of a square

2. Two charges of 1.0 $\mu C$ and 22.0 $\mu C$ are 0.50 m apart at two vertices of an equilateral triangle as in **Figure** 2 . [5]

    a. What is the electric potential due to the 1.0 $\mu C$ charge at the third vertex, point P?

    b. What is the electric potential due to the 22.0 $\mu C$ charge at P ?

    c. Find the total electric potential at P.

    d. What is the work required to move a 3.0 $\mu C$ charge from infinity to P ?

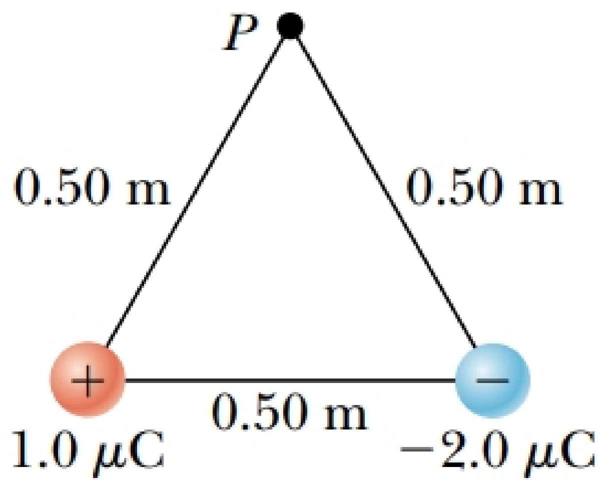


Figure 2:

3. A charge $+q$ is at the origin. A charge $-2q$ is at $x = 2.00m$ on the $x-axis$. For what finite value(s) of x is the electric potential zero. [5]

4. In Rutherford's famous scattering experiments that led to the planetary model of the atom, alpha particles (having charges of $+2e$ and masses of $6.64 \times 10^{-27}$ kg) were fired toward a gold nucleus with charge $+79e$ . An alpha particle, initially very far from the gold nucleus, is fired at $2.00 \times 10^{7}$ m/s directly toward the nucleus, as in **Figure** 3. How close does the alpha particle get to the gold nucleus before turning around? Assume the gold nucleus remains stationary. [8]

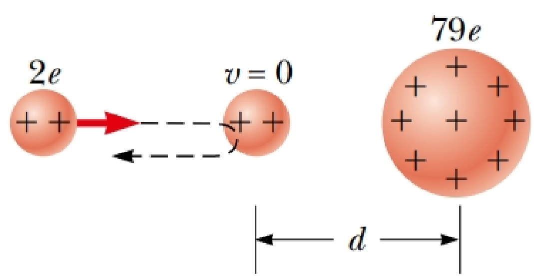


Figure 3:

5. The three charges in **Figure** 4 are at the vertices of an isosceles triangle. Calculate the electric potential at the midpoint of the base, taking $q = 7.00\,\mu C$ [8]

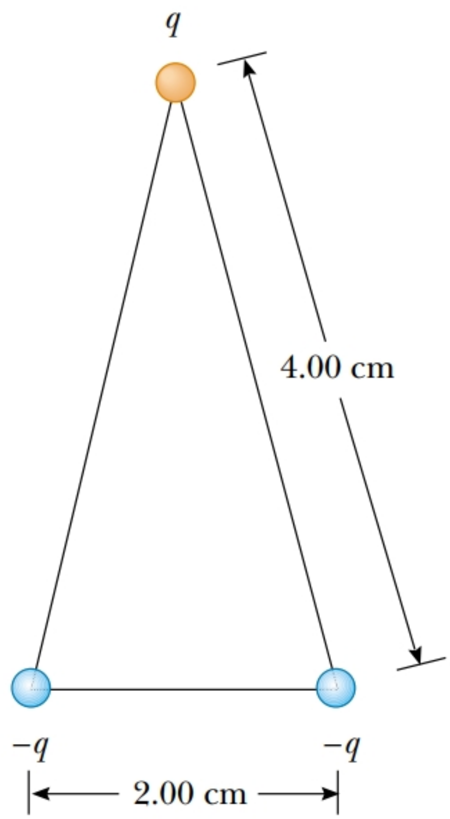


Figure 4:

6. Show that the amount of work required to assemble four identical point charges of magnitude Q at the corners of a square of side s is $5.41Ke\frac{Q^2}{s}$. [6]

***" Electric potential can be positive or negative like your mood and it's measured in Volts not Vodka! "***